

MA374 – Lab 06

(Rehan Sherawat)

Introduction

In this lab, stock price data from NSE and BSE for the period 2021–2025 is analysed. The objective is to study price behaviour at different time scales, understand statistical properties of returns, and simulate future price paths using Geometric Brownian Motion (GBM).

Daily, weekly and monthly prices are examined, followed by arithmetic and log return analysis. Finally, Monte-Carlo simulations are used to model possible price evolution in 2025.

Only representative assets and figures are shown since most stocks exhibit similar qualitative behaviour.

Representative assets:

- NSE Index (${}^N SEI$)
- RELIANCE (large cap)
- IDEA (small cap)

Weekly data uses the same weekday (Friday) and monthly data uses month-end prices.

—

Question 1: Price Behaviour

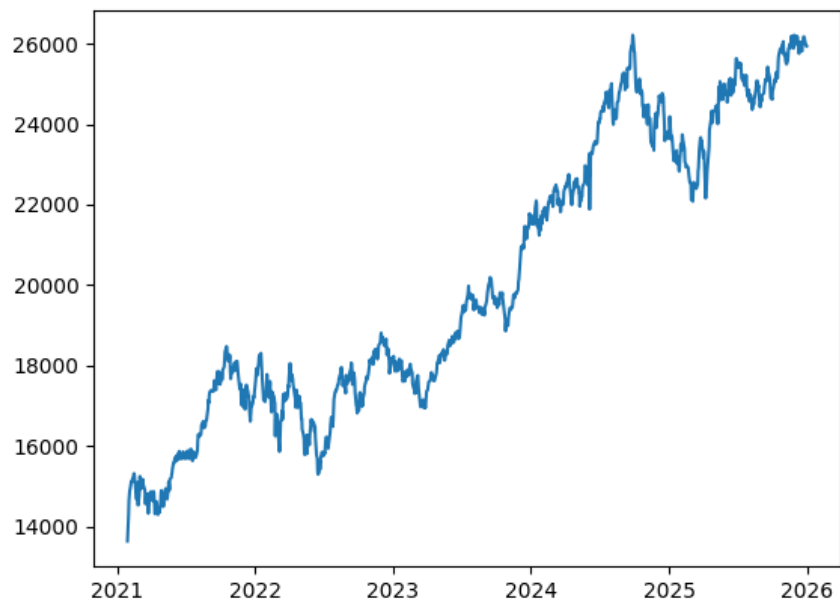


Figure 1: NSE Index Daily Prices

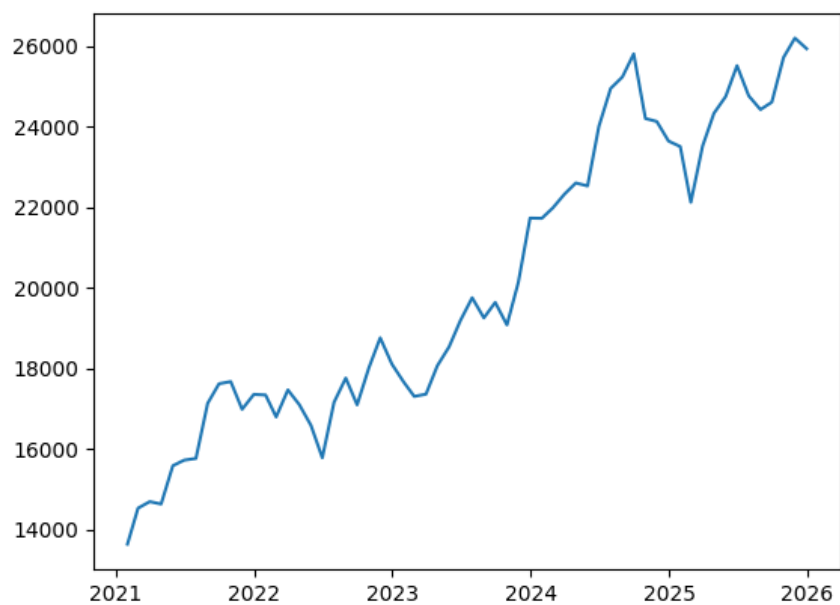


Figure 2: NSE Index Monthly Prices

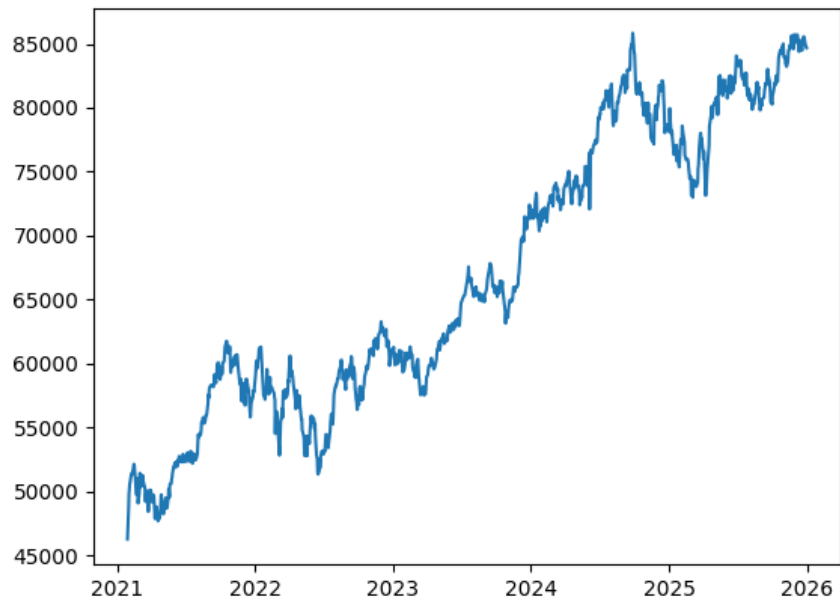


Figure 3: BSE Index Daily Prices

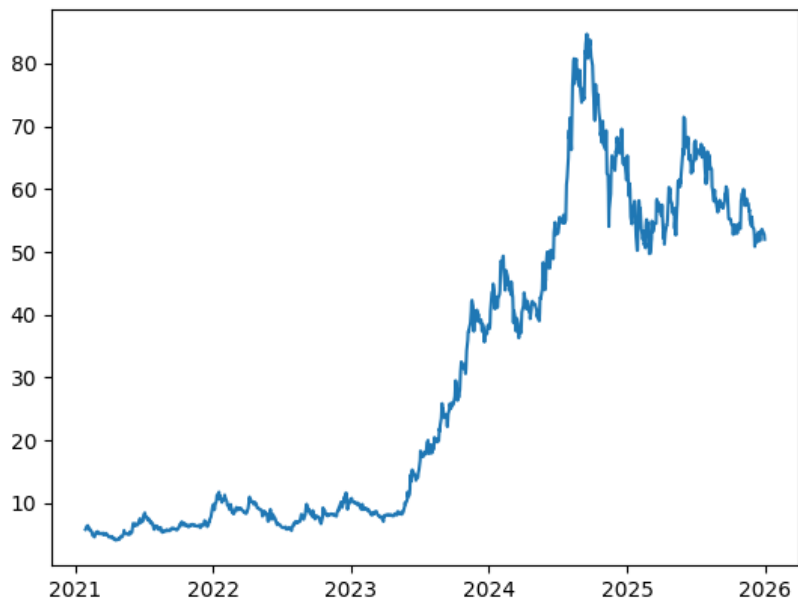


Figure 4: Small-cap Stock Daily Prices

Daily prices show strong short-term fluctuations due to frequent trading and news effects. Weekly aggregation reduces some noise, while monthly prices clearly highlight long-term trends.

Small-cap stocks display much sharper movements compared to indices and large-cap stocks, indicating higher volatility and sensitivity to market conditions.

Question 2: Arithmetic Returns

Arithmetic returns were computed and normalized to zero mean and unit variance.

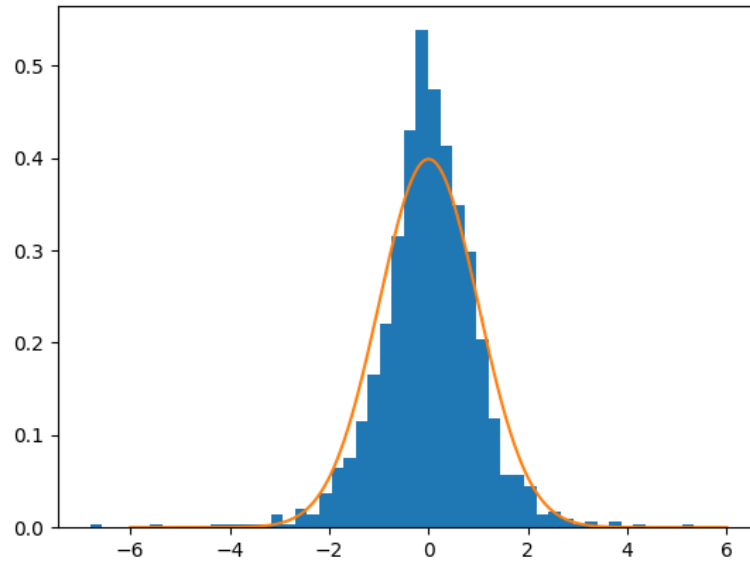


Figure 5: Normalized Daily Arithmetic Returns

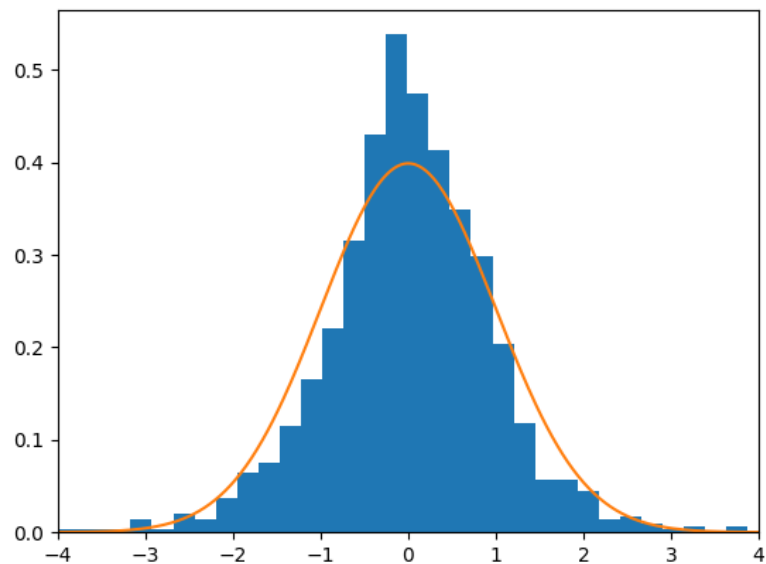


Figure 6: Tail Behaviour of Daily Arithmetic Returns

The central part of the distribution roughly follows a Gaussian shape. However, the tails are

much heavier, especially for daily returns, indicating a higher probability of extreme events than predicted by the normal distribution.

Monthly returns are closer to Gaussian, suggesting that aggregation over time reduces variability and extreme behaviour.

—

Question 3: Log Returns

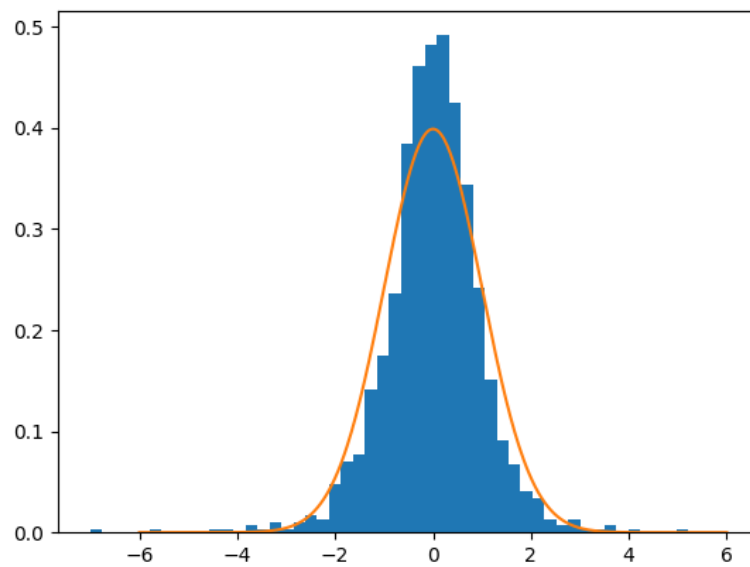


Figure 7: Normalized Daily Log Returns

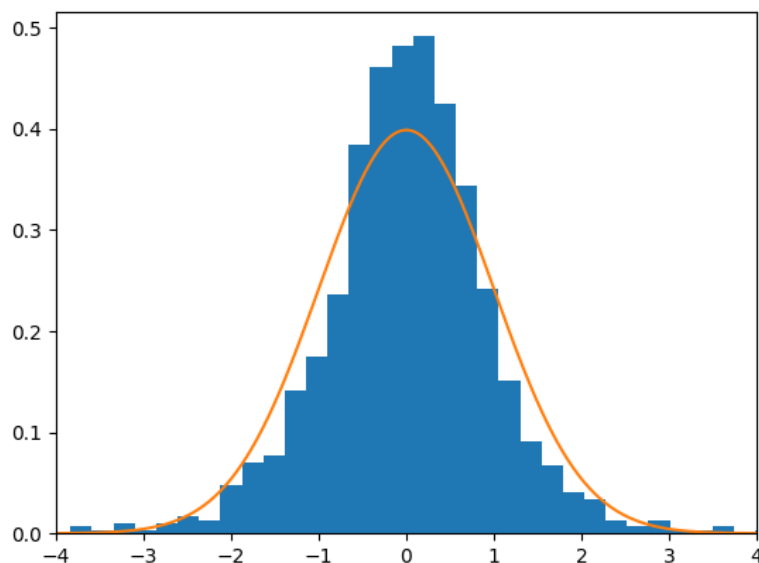


Figure 8: Tail Behaviour of Daily Log Returns

Log returns appear more symmetric compared to arithmetic returns. This is expected since log returns better handle multiplicative price changes.

Despite improved symmetry, fat tails persist, showing that financial returns deviate significantly from normality even after transformation.

—

Question 4: 2025 Price Simulation

Using log returns from 2021–2024, parameters μ and σ were estimated:

Asset	μ	σ
${}^N SEI$	0.000569	0.009113
RELIANCE.NS	0.000384	0.014604
IDEA.NS	-0.000356	0.037769
${}^B SESN$	0.000555	0.009191
RELIANCE.BO	0.000321	0.014954
IDEA.BO	-0.000368	0.037914

These parameters were used in GBM to simulate 2025 prices.

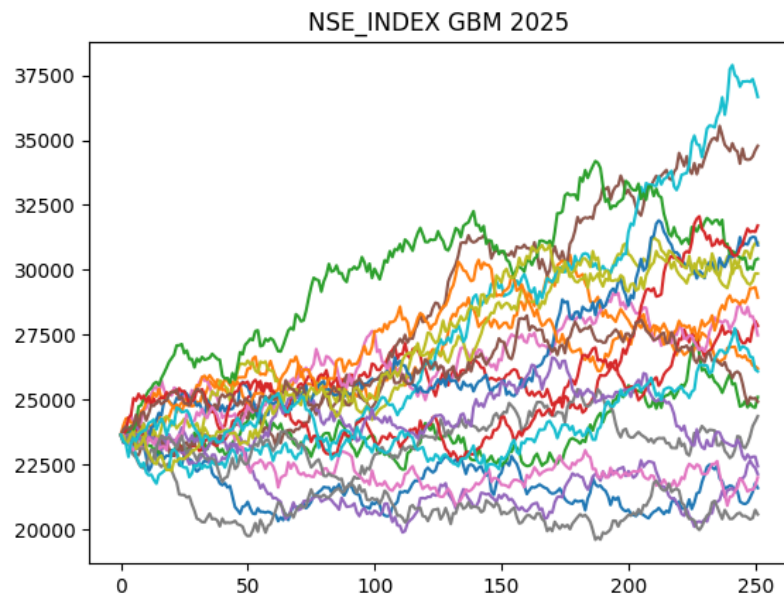


Figure 9: Simulated 2025 Paths – NSE Index

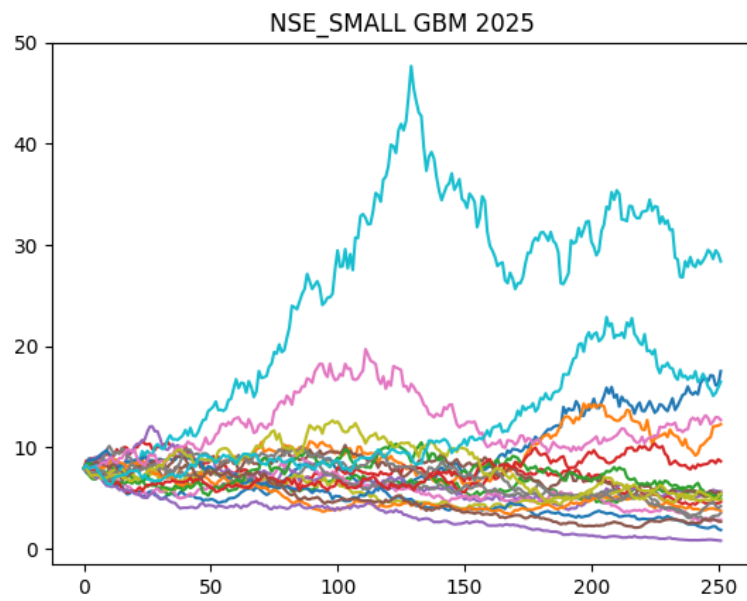


Figure 10: Simulated 2025 Paths – Small-cap Stock

Small-cap stocks show significantly wider simulated paths due to larger estimated volatility.

Question 5: Terminal Price Distribution

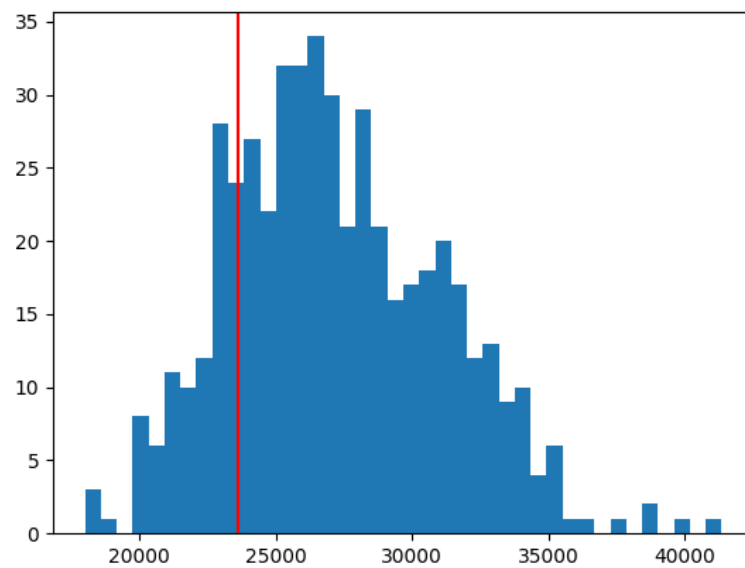


Figure 11: Terminal Price Distribution – NSE Index

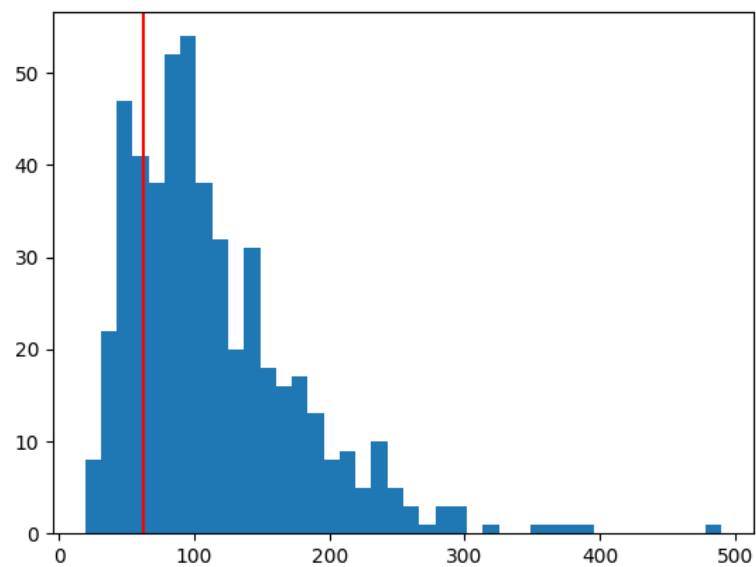


Figure 12: Terminal Price Distribution – Small-cap Stock

Terminal distributions for volatile stocks are much wider, indicating higher uncertainty in future prices. The red line represents the last observed price in 2024.

Overall Observations

- Daily returns exhibit heavier tails than monthly returns.
 - Log returns are more symmetric than arithmetic returns.
 - Small-cap stocks show higher volatility than large-cap stocks.
 - Simulated paths spread more for assets with larger σ .
 - Time aggregation improves normality and reduces noise.
-

Conclusion

This lab demonstrates key stylized facts of financial markets: heavy tails, volatility differences across assets, and deviations from normality at high frequencies. Log returns provide better symmetry, while aggregation over time reduces noise.

Monte-Carlo simulations using GBM highlight how volatility directly impacts uncertainty in future prices, especially for small-cap stocks. Overall, the analysis emphasizes the importance of volatility and return distribution properties in financial modelling.